

Challenge (Habanero)

- Q1.** A recipe for 24 cupcakes requires 15% of 2400 grams of flour. How many grams of flour are needed?
(A) 360
(B) 375
(C) 3600
(D) 369
(E) 390
- Q2.** A construction company has a 20% discount on all building materials. If a brick costs 25¢, how much will it cost after the discount?
(A) 18¢
(B) 20¢
(C) 22¢
(D) 25¢
(E) 30¢
- Q3.** A chef needs to reduce a recipe by 12% to serve 80% of the original number of guests. What percentage of the original recipe should be used?
(A) 68%
(B) 70%
(C) 72%
(D) 75%
(E) 88%
- Q4.** The length of a room is 15 meters and the width is 8 meters. If the length is increased by 25%, what is the new area as a percentage of the original area?
(A) 131%
(B) 135%
(C) 137.5%
(D) 140%
(E) 145%
- Q5.** A restaurant has 480 seats and is currently 60% full. What percentage of the restaurant's seating capacity needs to be filled to have 75% more customers?
(A) 87%
(B) 90%
(C) 93%
(D) 96%
(E) 99%
- Q6.** A builder needs to create a scale model of a building with a 30% scale reduction. If the original building is 120 meters tall, how tall is the scale model?
(A) 84 m
(B) 90 m
(C) 96 m
(D) 100 m
(E) 108 m
- Q7.** The price of a cake increases from 15¢ to 18¢. What is the percentage increase in price?
(A) 15%
(B) 16%
(C) 17%
(D) 20%
(E) 25%
- Q8.** A recipe calls for 0.5 liters of oil. If 25% of the oil is wasted, what percentage of the original amount of oil is left?
(A) 60%
(B) 65%
(C) 70%
(D) 72%
(E) 75%
- Q10.** A construction project has a budget of 10,000. If there is an increase of 8% in the cost of materials and then a 15% discount, what is the final cost as a percentage of the original budget? Explain your answer.

Open Ended Questions (FRQ)

- Q9.** A kitchen is currently 30% full of food. If 200 kilograms of food are added and then 25% of the total food is removed, what percentage of the kitchen's capacity is filled? (Show your work and calculations)

Chef's Tips

- Check the calculations step by step
- Review the units of measurements
- Ensure to apply the correct percentage operations